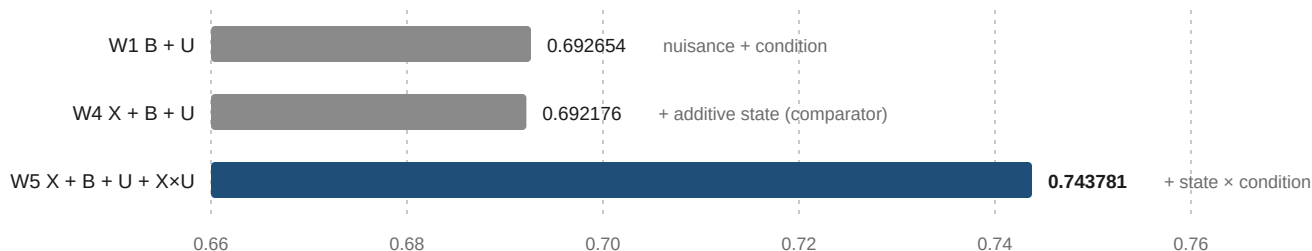


Figure 2 Preregistered clone-specific ranking result

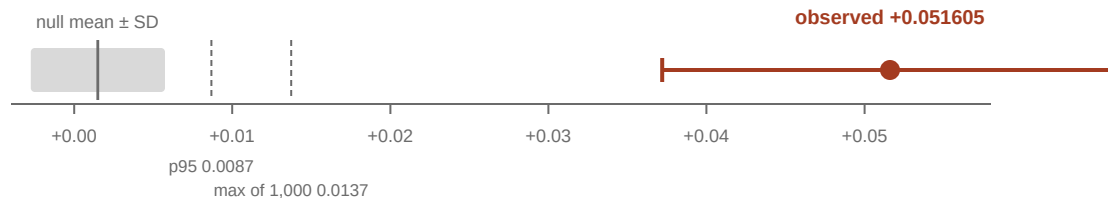
Equal-clone-weighted within-clone AUROC. Every degree of freedom fixed before these numbers existed.

A Ranking score by model



R(W4) sits BELOW R(W1): the additive state term adds nothing to ordering. The entire gain is the interaction.

B Observed Δ RANK against 1,000 full-refit permutations



0 of 1000 draws reached the observed value $\rightarrow p < 0.001$. 6.0 $\times$ the null p95; 11.8 SD above the null mean.

Permutation refits the entire pipeline inside every draw; observed-data hyperparameters are never reused.